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9.

	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
Diff	0.1	0.7	0.3	-0.1	0.5	0.2	0.5
Rank	1.5	7	4	-1.5	5.5	3	5.5

a. H_0：兩母體間無顯著差異

H_a：——〃——有——〃——

$T^+ = 1.5+7+4+5.5+3+5.5 = 26.5$

$T^- = 1.5$

$\min[T^+, T^-] = 1.5$

Critical Value ($n=7, \alpha=0.05$. 雙尾) $= 2 > 1.5 \Rightarrow$ Reject H_0

$\Rightarrow$ 兩母體間存在顯著差異.

b. $p\text{-value} = P(X \le 1) + P(X \ge 6) = 0.125$

$\because 0.125 > 0.1 \Rightarrow$ Non-Reject H_0.（不同於(a)）

$\Rightarrow$ Sign Test 僅利用差異的正負方向而未考慮幅度

11.

	1	2	3	4	5	6	7	8	9
A	3	14	7	10	9	6	13	6	7
B	7	12	9	15	12	6	12	5	13
diff	-4	2	-2	-5	-3	0	1	1	-6
Rank	-6	3.5	-3.5	-7	-5		1.5	1.5	-8

a. H_0: A. B 兩機器的月故障率無顯著差距

H_a: 〃 有 〃

$T^+ = 3.5 + 1.5 + 1.5 = 6.5$

$T^- = 6 + 3.5 + 7 + 5 + 8 = 29.5$

$\min[T^+, T^-] = 6.5$

Critical Value $(n = 8, \alpha = 0.05) = 4 < 6.5 \Rightarrow$ Non-Reject H_0

$\Rightarrow$ no sufficient evidence to indicate a difference in the monthly breakdown rate for two machines

b. 月產量，季節(溫度)，原料，保養狀態.

15. a. ① paired t-test, sign test, Wilcoxon signed-Rank Test

② 配對差值須來自一常態母體

③ 樣本過少，不能完美符合常態分布

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15 b. H_0: 兩母體無顯著差異

H_a: —〃— 有 —〃—

Sign Test

⊕ : 20 ⊖ : 0

$x = 20$

p-value $= 2 \times P(X \geq 20) = 2 \times C^{20}_{0} \times 0.5^{20} < 0.05$

$\Rightarrow$ Reject H_0

Wilcoxon sign Test

$T^+ = 210$, $T^- = 0$, $T = 0$

Critical value $= 52 > 0 \Rightarrow$ Reject H_0

c. 相同. 因為 With imagery 皆高於 without, 因此檢定皆能測出極度顯著的結果.

6.

8 13	9 16	4 2	4 2
14 20	7 10	6 7.5	7 10
9 16	5 5	7 10	5 5
12 19	8 13	8 13	5 5
10 18	9 16	6 7.5	4 2

$T_1 = 86$ $T_2 = 60$ $T_3 = 40$ $T_4 = 24$

H_0: 四種訓練時間與組裝時間無顯著差異. H_a: …… 有 ……

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{T_i^2}{n_i} - 3(N+1) = \frac{12}{20 \times 21} \times \frac{7396 + 3600 + 1600 + 576}{5} - 6 \times 21$$

$= 12.27$

$\chi^2_{3, 0.01} = 11.345 < 12.27 \Rightarrow$ Reject H_0

$\Rightarrow$ 不同訓練時間對組裝時間有顯著差異

8. a. H_0：四組母體分布相同，H_a：不相同

$T_1 = 247.5 \quad T_2 = 168, \quad T_3 = 216.5, \quad T_4 = 188$

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{T_i^2}{n_i} - 3(N+1) = \frac{12}{110} \cdot \frac{247.5^2 + 168^2 + 216.5^2 + 188^2}{10} - 3 \times 11$$

$= 2.632$

$df = k-1 = 3$

$\chi^2_{3,0.01} = 11.345 > 2.632 \Rightarrow$ Non-Reject H_0

b. $\chi^2_{3,0.1} = 6.251$

$\Rightarrow$ p-value > 0.1

$\Rightarrow$ p-value ≈ 0.452

c. ANOVA Test：p-value $= 0.468$

H-test：p-value $= 0.452$ $\Rightarrow$ 皆 non-Reject H_0

Implication：兩者結果相同 $\Rightarrow \begin{cases} \text{數據可能符合常態} \\ \text{Robustness} \end{cases}$